

DUSTRUNNER · REQUIREMENTS

DustRunner requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/dustrunner		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a reference row and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

DustRunner requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users or a pilot site, and will be checked by calculation at TRL 3. The status column compares them with the first-order estimates in DRN-PRC-001. Three requirements are **not met or not yet shown to be met** on paper: R14 (payback) is not met in the mid case, and R2 (cleaning effectiveness) and R3 (glass safety) cannot be shown without tests.

The **reference row** used throughout is one fixed-tilt table, one module high in portrait (1P), with 35 modules of 2,278 x 1,134 mm and about 580 W each: about 40 m long, about 90 m² of glass and about 20 kWp, tilted 25°, with the lower glass edge about 0.6 m above the ground.

ID	Requirement	Target	Status (concept, estimate)	Verification (TRL 3 or later)
R1	Water-free cleaning	No water or liquid used in normal operation	Met by design	Design review
R2	Cleaning effectiveness on dry dust	Daily dry pass keeps the average soiling loss at 1.5 % or less when the site soiling rate is 0.3 to 0.5 % per day	Not shown ; depends on dust type and dew	Literature review at TRL 3; later soiling-station comparison
R3	Glass and coating safety	Only microfiber touches the glass; brush interference 3 to 5 mm; no hard bristles, grit traps or metal within 10 mm of the glass; complies with the module maker's cleaning guidance	Not shown ; abrasion over years is unknown	Maker guidance review; later abrasion test on sample glass
R4	Fit the reference table	1P portrait modules 2,278 mm long with 30 to 35 mm frames; short (top and bottom) edges free of clamps; tilt 10 to 35°; gaps between modules up to 25 mm; frame height steps up to 5 mm	Met on paper for the reference table; other module lengths need the beam and brush cut to length	Survey of target tables; model check
R5	Coverage	95 % or more of the glass area brushed per pass	About 98 % (2,198 of 2,228 mm of glass across the slope)	Model check
R6	Rail-free installation	No rails along the row; only a dock at one end and clamp-on end stops at the other; no drilling of modules or structure; installed by two people in 60 min or less	Met by design; time unverified	Installation sequence review
R7	Cycle time	A 100 m row cleaned out and back in 20 min or less	About 17 min at 0.2 m/s	Calculation
R8	Energy autonomy	Three daily cycles on a 100 m row with no sun; recharged from the dock panel in one day at 3 peak sun hours	About 102 Wh usable against about 58 Wh for three cycles; about 45 Wh harvested at 3 sun hours against about 20 Wh per cycle	Energy calculation
R9	Stay on the row	Two independent means stop the robot at each row end; hook rollers keep it on the table in operating wind; operation only below 8 m/s wind; robot parked and latched in the dock survives 35 m/s	End stops met by design; wind limits unverified	Wind load calculation; design review
R10	Load on the modules	Robot mass 15 kg or less; each wheel load 60 N or less and carried on the frame only; no load on the glass except brush contact	About 12 kg; about 52 N per wheel with hook preload	Mass estimate; module maker's guidance
R11	Environment	Electronics IP65; operates at 0 to 50 °C ambient on glass up to 75 °C; UV-stable plastics; dust-sealed bearings	By design; unverified	Datasheets and design review
R12	Safe to be near	Emergency stop button on each end truck; brush and drives stop within 2 s of a stop	By design; unverified	Design review; later bench test

ID	Requirement	Target	Status (concept, estimate)	Verification (TRL 3 or later)
		command or of a truck lifting off the frame; brush ends guarded by the hood; nothing on the robot can reach PV cables, connectors or junction boxes		
R13	Affordable	Robot and dock, one row, \$500 or less in parts	About \$485 (indicative), thin margin	Priced BOM (bom/ bom.csv)
R14	Pays for itself	Value of the recovered energy covers the parts cost in 3 years or less on the reference row, mid case (0.3 % per day soiling, monthly manual wash as baseline, \$0.10/kWh)	Not met: about 4 years in the mid case; about 2 years at 0.49 % per day or on a 100 m row	Energy yield calculation
R15	Serviceable	Microfiber sleeve replaced in 15 min or less without taking the robot off the row; motors, bearings, wheels and battery are standard catalog parts	By design	Design review

■ Assumptions

- Specific yield about 5.5 kWh per kWp per day for a clean array at a sunny desert site (estimate).
- Soiling rate 0.3 % per day (mid case) and 0.49 % per day (high case, as measured at QEERI in Qatar; see DRN-PRB-001).
- Baseline practice is a manual wash every 30 days. Longer manual intervals, or no cleaning at all, make DustRunner's case stronger.
- Electricity or displaced diesel valued at \$0.10/kWh. Diesel-displacing mini-grids are often higher.
- 20 W dock panel, 75 % derating for heat, dust, wiring and a simple charge controller.
- Wind and structural values are first-order and will be checked at TRL 3.